DOMINATION OF MODULE MAPS CAN BE TESTED ON FINITELY PRESENTED MODULES

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Allowed standard prerequisites (editorial).

- Every module is a directed colimit of finitely presented modules. Used at the start of the counted proof (lines 21143-21145). Replaces an arbitrary test module by a directed system of finitely presented ones.
- Exactness of directed colimits and their commutation with tensor products. Used at the end of the counted proof (lines 21147-21149). Passes the kernel inclusion from each term of the system to the colimit.

Lemma 0.1. *Let $f : M \to N$ and $g : M \to M'$ be maps of R-modules. Then g dominates f if and only if for any finitely presented R-module Q, we have* $\mathrm{Ker}(f \otimes_R id_Q) \subset \mathrm{Ker}(g \otimes_R id_Q)$.

Proof. Suppose $\mathrm{Ker}(f \otimes_R \mathrm{id}_Q) \subset \mathrm{Ker}(g \otimes_R \mathrm{id}_Q)$ for all finitely presented modules Q. If Q is an arbitrary module, write $Q = \mathrm{colim}_{i \in I} Q_i$ as a colimit of a directed system of finitely presented modules Q_i. Then $\mathrm{Ker}(f \otimes_R \mathrm{id}_{Q_i}) \subset \mathrm{Ker}(g \otimes_R \mathrm{id}_{Q_i})$ for all i. Since taking directed colimits is exact and commutes with tensor product, it follows that $\mathrm{Ker}(f \otimes_R \mathrm{id}_Q) \subset \mathrm{Ker}(g \otimes_R \mathrm{id}_Q)$. □

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